

**Guideline Title: General Obstetric Patient Assessment**

**Guideline Number: GUID-3203-M022**

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| Issue date: 7/1/2024                                                                         | Replaces Dept. Policy:      |
| Revision dates: 4/23/24                                                                      | Developed by: Air Care Team |
| Approved by: Dr. Christopher Hunter, MD<br>Air Care Team Medical Director                    | Approved by:                |
| Signature:  | Signature:                  |
| Department Numbers: 3203                                                                     |                             |

- I. **PURPOSE:** To provide guidelines for the assessment, treatment, and transport of a pregnant patient, including those in active labor and potential complications of pregnancy. The goal is to optimize the outcome for the mother and baby by minimizing the risk of in-flight deliveries.
- II. **Considerations for transport**
  - a. For purposes of safety, the transport team will consider a viable pregnancy as any pregnancy over 20 weeks gestation based on the best gestational age available.
  - b. In the case of a pregnant woman with an Emergency Medical Condition, the woman is considered stabilized when she has delivered the child and placenta. Until then, the woman and child are considered unstable.
- III. **DEFINITIONS:**
  - a. Active Labor (Medical Definition): The period when the cervix dilates from four to eight centimeters. Contractions usually come between three and five minutes apart, dilating the cervix approximately one centimeter per hour.
  - b. Active Labor (EMTALA Definition): EMTALA defines “active labor” as labor at a time when
    - i. There is inadequate time to effect safe transfer to another hospital prior to delivery
    - ii. A transfer may pose a threat to the health and safety of the patient or the unborn child.
    - iii. This statutory definition renders irrelevant any medical definition of active labor. It has retrospective implications (e.g., if a woman delivers during transport), she was previously in active labor.
- IV. **EQUIPMENT**
- V. **DEPARTMENT GUIDELINE:**
  - a. Establish care as defined by *GUID-3203-G002 – General Patient Care*.
  - b. Pregnancy & Trauma:
    - i. Transport all pregnant patients meeting Trauma Alert or Trauma RED criteria to a State Approved Trauma Center
    - ii. Trauma Yellow or Green can be transported to the EMERGENCY DEPARTMENT of closest OB receiving facility
    - iii. Treat per *GUID-3203-T017 Trauma in Pregnancy*

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- c. Obtain maternal baseline vital signs and Fetal Heart Rate.
- d. Obtain History from sending facility as soon as possible. This must include the Estimated Gestational Age (EGA) based on last menstrual period or recent ultrasound and Estimated Date of Confinement (EDC). It must also include information about contractions:
  - i. Onset.
  - ii. Frequency.
  - iii. Intensity - mild, moderate, strong.
  - iv. Duration - timed from the moment uterus begins to tighten until it relaxes.
- e. Obtain maternal history:
  - i. Non-OB: Allergies, medical/surgical history, medications.
  - ii. OB History: Pre-natal care, presence of more than one fetus, placenta previa, history of pre-term labor (PTL), any complications with this pregnancy, rupture of membranes, presence of meconium in amniotic fluid, recent exposure to infections, blood type and Rh, time and amount of last oral intake, vaginal bleeding, fetal activity/movement felt, etc.
  - iii. Previous complications with other pregnancies.
  - iv. High-risk conditions (e.g., pre-eclampsia, abruption, placenta previa, etc.).
  - v. Gravida & Parity: G# (number of pregnancies); P# (number of pregnancies carried to > 20 weeks. Also expressed as G # P# # # #. For parity, numbers stand for term births, premature births, abortions, and living children (in that order).
  - vi. Number of previous cesarean sections.
- f. The transport team is obligated to ask the sending facility/physician for the results of the medical screening exam which will aid in the diagnosis of an emergency medical condition. This should include the time of the last cervical check, effacement (shortening and thinning of the cervix expressed as a %), dilation (enlargement of cervical opening, anywhere from 1 cm to 10 cm) and station (presenting part of fetus: # cm above (-) or below (+) an imaginary line between maternal Ischial spines (birth is +4 to +5) of the pregnancy. Because of the changing nature of conditions in pregnancy, it may be necessary to ask for a repeat pelvic exam if not recently done.
- g. The medical transport team does not perform pelvic exams or cervical checks. This information should be obtained from the sending physician. When indicated, vaginal examination should be done at referring facility within 15 minutes of team arrival by referring physician or repeated before you leave.
- h. **Consult with Medical Direction**
  - i. If delivery is *imminent*, advisability of maternal transport should be questioned, and ACT should contact the receiving physician and/or medical control.
  - ii. If significant possibility of an in-transport delivery exists, consult with the receiving physician and/or medical control (conferencing an obstetrical specialty physician medical director as needed). Consultation should be sought to aid in decision-making regarding transport vs. delivery at the sending facility. In general, transport teams should err on the side of caution before transporting such patients as the delivery of the infant in the sending facility is almost always in the best interest of the patient and unborn child when compared to an in-transport delivery.

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- iii. The following patients are *not* recommended for transport without a consultation with the receiving physician and/or medical control (conferencing an obstetrical specialty medical director as needed)
      - 1. The mother in active labor whose cervix is dilated more than 5 cm. It should be noted that rapid progression of labor can occur, especially in multiparous mothers, so the flight team and the physicians involved may choose to be more conservative than this in some patients in active labor.
      - 2. The mother in active labor who is less than 37 weeks gestation with ruptured membranes.
      - 3. The mother in active labor whose membranes are ruptured and meconium staining is present.
      - 4. Contact the receiving physician and/or physician medical director (conferencing an obstetrical specialty physician medical director as needed) with any fetal heart rates under 100 beats per minute.
    - iv. In general, aircraft should not be used to transport patients in active labor unless there are special circumstances. Ground transport should be utilized unless directed by medical control.
  - i. General Transport
    - i. Preserve patient's warmth and privacy.
    - ii. Confirm that appropriately sized resuscitation equipment (for both mother and infant) is available and functioning.
    - iii. Secure the pregnant patient by positioning the safety straps low on the pelvic girdle. Straps across the abdomen can lead to indirect trauma to the fetus in the event of significant turbulence or decelerations forces.
    - iv. If possible, ensure patient has empty bladder prior to departure.
    - v. If delivery occurs during a flight, every attempt should be made to land the aircraft at the nearest emergency department.
  - j. Interventions
    - i. If patient has adequate spontaneous respirations, administer supplemental oxygen to maintain SpO<sub>2</sub> > 93%.
    - ii. Connect to cardiac monitor, NIBP, pulse oximeter and ETCO<sub>2</sub> (when indicated to maintain ETCO<sub>2</sub> 35 - 45 mmHg).
    - iii. Obtain a set of vital signs and repeat every 10 minutes, or more frequently depending on presentation.
    - iv. Establish IV access x 2 and consider blood draw for lab studies.
    - v. Consider maintenance fluids of 0.9% NS/LR at 100 - 150 mL/hr IV.
    - vi. Consider use of blood products for ongoing hemorrhage and hemodynamic instability per GUID-PR002 - *Blood Product Administration* or GUID-PR025 - *Emergency Blood Product Administration*. Observe the patient for signs and symptoms of disseminated intravascular coagulopathy.

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- vii. If the patient has a GCS of < 9 and/or has a deteriorating level of consciousness, consider advanced airway management.
- viii. Nauseated patients may be treated with anti-emetics per *GUID-3203-G004 Nausea Vomiting Motion Sickness*
- ix. Consider NG or OG tube for gastric decompression per *GUID-3203-PR010 Gastric Tube Insertion*
- x. Treat pain and anxiety per *GUID-3203-G005 Analgesia and Sedation Management* and *GUID-3203-G006 Anxiety- Combative Patient Management*
  - 1. Use cautiously if delivery is imminent secondary to fetal respiratory depression.
  - 2. Report to the receiving unit when opioids are administered to the patient.
- xi. Prolapsed Umbilical Cord
  - 1. Elevate mother's hips and place in knee to chest position
  - 2. Encourage mother not to push
  - 3. Evaluate fetal heart rate/cord pulsation
  - 4. Cover cord with moist sterile dressing
  - 5. Elevate pressure off cord, do not attempt to push the cord back into vagina
- k. Pre-term labor
  - i. Consider insertion of indwelling urinary catheter to ensure bladder emptying as bladder distension may induce uterine contractions.
  - ii. Consider group B strep prophylaxis; discuss with the receiving physician and/or medical control to determine appropriate treatment.
  - iii. Consider steroid administration with **Betamethasone** 12 mg IM for lung maturity for gestational age < 37 weeks; discuss with the receiving physician
  - iv. Consult the receiving physician and/or medical director to consider tocolytic therapy
    - 1. **Terbutaline Sulfate 0.25 mg SQ** every 15 - 30 minutes up to a maximum of 1 mg  
**OR Terbutaline infusion** 5 mcg/min IV to a maximum 80 mcg/min IV
      - a. Contraindications to Terbutaline Sulfate: cardiac history, maternal HR > 140, maternal BP < 90/60 mmHg, fetal heart rate > 160, signs of maternal pulmonary edema.
      - b. If the patient is on a Terbutaline infusion at the referring facility, contact the receiving OB physician regarding their management.
    - 2. **Magnesium Sulfate:**
      - a. Infuse 4 g IV bolus over 20 - 30 minutes, followed by 2 - 4 g/hr IV infusion.
      - b. Patients receiving Magnesium Sulfate infusion must be closely monitored for toxicity (somnolence, muscular paralysis, respiratory depression, hypotension, or loss of DTRs)
        - i. Immediately discontinue Magnesium Sulfate infusion.
        - ii. Administer **Calcium Gluconate 1 g** of 10% solution over 1 - 2 minutes.
- l. Post-partum hemorrhage
  - i. Ensure 2 large bore IVs are patent.

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- ii. For atonic uterus:
  - 1. Firmly massage uterine fundus.
  - 2. Continue **Oxytocin** infusion from receiving facility (General recommendation 20 units/1,000 mL NS or LR at 20 - 40 milliunits/min (60 - 120 mL/hr) or if no IV access, Oxytocin 10 units IM)
- iii. For visible lacerations, apply direct pressure.
- iv. Administer **TXA** per *GUID-3203-T016 - Tranexamic Acid for Severe Hemorrhage*.
- v. If hypotensive, place patient in the left or right lateral recumbent position and treat per *GUID-3203-M011 - Hypotension*.
- vi. For hemorrhagic shock that does not respond to warmed fluid resuscitation, administer blood products according to *GUID-3203-PR002 - Blood Product Administration* or *GUID-3203-PR025 - Emergency Blood Product Administration*.
- m. Eclampsia and Pre-Eclampsia
  - i. Definitive treatment can only be accomplished through delivery of the fetus(es).
  - ii. Seizure prevention should be accomplished through administration of **Magnesium Sulfate**
    - 1. **Magnesium Sulfate:**
      - a. Infuse 4 g IV bolus over 20 - 30 minutes, followed by 2 - 4 g/hr IV infusion.
      - b. Patients receiving Magnesium Sulfate infusion must be closely monitored for toxicity (somnolence, muscular paralysis, respiratory depression, hypotension, or loss of DTRs)
        - i. Immediately discontinue Magnesium Sulfate infusion.
        - ii. Administer **Calcium Gluconate 1 g** of 10% solution over 1 - 2 minutes.
  - iii. If SBP > 160 mmHg or DBP is > 110 mmHg, administer:
    - 1. **Labetalol 20 mg slow IV push** (over 2 min). May repeat every 10 minutes with additional doses of 40 mg and then 80 mg, and then may repeat the 80 mg x 2 for the maximum total dose of 300 mg until DBP <100.
    - 2. If BP is not responsive to Labetalol then obtain from receiving facility and administer **Hydralazine 5 mg IV** repeated as needed every 20 minutes at a dose of 5 - 10 mg IV to a total dose of 40 mg.
  - iv. Seizure activity should be treated with:
    - 1. Supportive care first (maintain airway, administer supplemental oxygen as needed to maintain SpO2 > 93%, prevent injury).
    - 2. As a first line treatment, administer **Magnesium Sulfate 4 g IV bolus**, then an infusion of 2 g/hr IV.
    - 3. For seizure lasting greater than 2 minutes, administer
      - a. **Midazolam 5 mg IV/IN or 10 mg IM**; repeat as necessary every 2 - 5 minutes.
      - b. For persistent seizures continue Magnesium Sulfate infusion and treat per *GUID-3203-M015 Seizures*

**VI. DOCUMENTATION: EMS Charts**

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**VII. REFERENCES:**